Document Effective Date
08/09/2024

قطر للطاقة
QatarEnergy

IS LIFTING EQUIPMENT CONTROL AND OPERATIONS PROCEDURE

DOC NO: IS-HSE-PRC-027

Rev: 03

Quality Reviewed By: OIB/4 Digitally signed by Moklis Miah Date: 2024.09.05 13:11:28 +03'00'

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	OIH/12	A/OIH/1

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1.0 PURPOSE

This procedure defines requirements and roles & responsibilities of personnel involved in the supply, operation and control of lifting equipment in QatarEnergy IS lifting and cargo handling operations in addition to those defined in CORP-ENG-PRC-038 (Corporate Procedure for Lifting Equipment and Operations) and QP-PAI-STD-005 (Corporate Standard for Lifting Equipment and Operations). In case of conflict between these corporate documents and subject procedure, the more stringent requirements shall be applicable.

2.0 SCOPE AND APPLICABILITY

This Procedure covers all QatarEnergy-IS operated worksites. It applies to all employees of QatarEnergy-IS including employees of any contractor or outside agency that work at any facilities owned, operated and / or managed by QatarEnergy-IS. Contractor companies may operate to their own standards only if it can be demonstrated that they are equal to or exceed QatarEnergy-IS minimum operating standards.

3.0 ROLES AND RESPONSIBILITIES

PERSON IN CHARGE (PIC)

The Person in Charge is the person who has responsibility for his installation. The Person in Charge shall be one of the following:

- Offshore Installation Superintendent (OIS) in case of PS-1
- Offshore Installation Manager (OIM) in case of a drilling rig
- Barge Master in case of a Lift boat / Barge
- Captain of workboat or vessel
- Contractors Senior Worksite Supervisor.

The Person in Charge is responsible for:

- Organize Examinations on both fixed and loose lifting equipment at the worksite.
- Nominate a suitably experienced/knowledgeable person to act as the installation's on-board lifting operations focal point.

RIGGING SUPERVISOR

- Ensure crews are aware and familiar with current Lifting Advisory.
- Ensure crews are aware and familiar with QatarEnergy Lifting requirements.
- Ensure that crews have relevant and up to date training.
- Ensure all Lifting Operations on site are carried out in accordance with QatarEnergy requirements.
- Review and approve all lifting operations, lifting plans and risk assessments.
- Act as lifting competent person on site and be able to develop effective Lifting Plans.
- Determine suitable levels of Supervision for complex lifts.
- Ensure all personnel involved in lifting operations meet the minimum competence standards
- Ensure all lifting equipment used is fully certified and fit for service.

OFFSHORE INSTALLATION SUPERINTENDENT (OIS)

- Stationed on the offshore installation and has the overall responsibility for activities in the respective field, including the safe transfer of personnel from / to marine vessels at offshore installations and facilities.

OFFSHORE DRILLING SUPERVISOR (ODS) / BARGE SUPERINTENDENT

- Ensure the safety of drilling, well servicing or pipe-lay- barge operations, as applicable.
- Approve transfers of personnel to / from your facility.

OFFSHORE CONSTRUCTION SUPERVISOR (OCS)

- Ensure the safety of construction operations on wellhead jackets, satellites and tripods.
- Approve all transfers of personnel to / from the facility on which you are supervising construction operations.

MARINE VESSEL CAPTAIN

- Ensure the safety of all personnel onboard his marine vessel and that his deck crew are fully briefed.
- Approve all transfers with prior agreement with the OIS or designate of personnel to / from the marine vessel.
- Discuss all proposed personnel transfers with the appropriate personnel at the facility (OIS or HSE rep), prior to conducting the personnel transfer.
- Be in radio contact with the crane operator and with personnel supervising the transfer of personnel on the deck of the marine vessel and on the facility.
- Alert the nearest standby marine vessel equipped with an FRC and also the radio operator, if any person riding a personnel basket / device should fall into the sea.

PERSONNEL TRANSFER SUPERVISOR (OFFSHORE FACILITY) / PERSONNEL TRANSFER SUPERVISOR (MARINE VESSEL) AD-HOC

- Discuss all proposed personnel transfers with the marine vessel captain or with the facility supervisor, as appropriate, prior to conducting any transfer of personnel using the personnel transfer basket / device.
- Be in radio contact with the vessel captain, with the crane operator and with each other.
- Have man overboard life rings immediately available for emergency use in a man overboard situation.
- Be present at the scene of all transfers of workers via personnel basket / device and shall supervise the safe loading and unloading of personnel.
- Provide on-site instruction to any workers who appear unfamiliar with the Procedure used in riding the personnel basket / device.

CRANE OPERATOR (ALL LOCATIONS)

- Discuss all proposed personnel transfers with appropriate personnel at the facility and with the vessel captain, prior to conducting the personnel transfer.

- Be in radio contact with the vessel captain and with those individuals supervising the transfer of personnel on the facility and on the deck of the marine vessel and shall have full view of the banksman and/or the pickup and set down areas.
- Abort the personnel transfer operation and return the personnel transfer basket / device to its point of origin, if it is unsafe to continue the transfer due to weather or other factors.

MATERIALS MAN (AS APPLICABLE FOR AND OTHER VESSELS IN QATARENERGY-IS FIELDS)

- Check the lifting equipment being used is in good condition, certified for use, correctly color coded and of sufficient capacity to carry out the lift.
- Ensure both the, Banksman, Vessel Master and the crane operators are familiar with the method or communication of signalling (both radio and visual) to be used and has a full view of the transfer areas.
- Ensure that the Banksman and Deck Supervisor are clearly identifiable.

4.0 PROCEDURAL STEPS

Deviation from the requirements of this procedure may only be given under special circumstances. If such an occasion arises, the person in charge of the lifting operation must request dispensation. These situations are dealt with on a case by case basis and deviation will only be granted in an emergency or under special circumstances and when the operation has been further risk assessed, provided that the hazards are assessed and approval is granted by the Proponent Manager or designate and HSE Manager or designate.

4.1 LIFTING AND CARGO HANDLING OPERATIONS

All cargo will be pre-slung at the shipping base and, placed in the appropriately sized cargo container, suitably secured to minimize the amount of handling by the supply vessel crew. The weight of each individual cargo item will be recorded on the shipping manifest.

There are also fundamental rules that are mandatory and must be adhered to, to ensure safe operations. They are as follows:

1. A plan and risk assessment has been completed and the lift method and equipment has been determined by a competent person(s).
2. Operators of powered (or manual) lifting devices are trained and certified for that equipment.
3. Rigging of the load is carried out by a competent person(s).
4. Lifting appliances and accessories have been certified for use within the last 6 months by a competent inspector and display the current lifting color code.
5. Load weight is known and does not exceed dynamic and / or static capacities of the lifting equipment.
6. Any safety devices installed on lifting equipment are operational.
7. All lifting devices and equipment have been visually examined before each lift by the user.
8. A clear and effective communication system is employed and understood by all personnel involved with the lifting operation. Note: All offshore cranes used for vessel operations shall be equipped with a fixed, hands free operated VHF radio.
9. An Appropriately Sized Safety pennant is attached to the cranes block.
10. All personnel must keep out of any area where they may be injured by a falling or shifting load. No personnel may stand below, return below, or stand on top of a suspended load.

All personnel involved in the lifting operation must ensure a route of escape and never stand between a load and a wall / container / barrier etc.

11. Immediately a lift deviates from the plan or an unexpected complication arises, the lifting operation must be stopped, made safe and reassessed. All personnel must remain clear of the lift until reassessment / re-planning of the lift has been carried out.
12. Lifting operations involving cranes will be undertaken by a minimum of three competent & trained personnel: crane operator, banksman and load handler.
13. The crane operator is responsible while the load is in the air. The banksman controls the initial lifting of the load, laydown of the load and lifts that are out of the line of vision of the crane operator.
14. The banksman shall:

   - Ensure that he is easily identifiable from other personnel by wearing a hi-viz jacket or waistcoat with reflective tape, which is clearly marked to indicate that he is the authorized crane banksman.
   - Not touch the load and stand back from the load being handled in a prominent position where he has a good view of the lifting activities.
   - Remain in communication with the load handler and crane operator at all times, i.e. primary communications should be in line of sight using hand signals.

15. For cargo operations on static locations the load handler (s) shall:

   - Stand clear while a load is lifted clear of the deck and landed, while slack is taken up with or without a load on the hook and must confirm to the banksman that he is clear.
   - Not touch a load being landed until it is below waist height and never attempt to manually stop a swinging load. In some essential laydown areas, it is necessary to maneuver containers into limited landing areas adjacent to handrails, where the handrails are slightly in excess of waist height.
   - Be easily identifiable, and distinct from the banksman.

16. For all vessels securely tied up to a berth at RLIC, this is considered a static operation. Load handlers may touch the load providing the load is not swinging the load is under waist height, no part of their body is under the load and an escape route is available. However, Hands off is the preferred method of cargo handling.
17. For QatarEnergy-IS operated installations and onshore sites stacking of containers, baskets, tanks and half heights should be discouraged unless:

   - Equipment is specifically designed for that purpose, and clearly marked as suitable for stacking.
   - No stack shall be more than two units in height.
   - Stacked containers must be able to lower the crane hook to the load handler so that the crane hook can be attached/detached while the load handler is standing at deck level.

18. QatarEnergy-IS Hands off Policy.
   QatarEnergy-IS operates strict hands-off policy for supply vessel crews. Load handlers will not touch loads at any time and will remain clear in a safe position away from the intended path of the load, until the load is secure in its final position on the deck of the vessel and the rigging has been lowered to be unhooked.

For vessel to platform lifts, load handlers may touch the load, providing the load is not swinging, the load is under waist height, no part of their body is under the load, weather conditions permit, an escape route is available and risk assessment of task determines that hands on is required to safely orientate the load on the platform.

Crane operators will not raise or lower a load from / to a supply vessel until all crew members are clear of the area and in a safe position away from the intended path of the load.

Containers, CCUs, Waste Skips, Small Baskets – The use of tag lines should be avoided as they could involve personnel standing in unsafe positions. If it is considered necessary, they shall only be used after a Risk Assessment has been undertaken.

Long baskets, Tubulars, Fragile items of cargo, Heavy Lifts – When being offloaded from a supply vessel, taglines will be installed. When being back loaded, if it is considered necessary, and weather conditions permit, taglines shall only be used after a risk assessment has been undertaken, if the master of the vessel agrees and the following precautions are adhered to as a minimum:

- Taglines must be made up from a single continuous length of rope of appropriate diameter for the weight of the load.
- Apart from the knot attaching the line to the cargo, there must be no other joints or knots in the line.
- Tag lines must be of sufficient length to allow personnel handling cargo to work at a safe distance well clear of the immediate vicinity of the load.
- All sections of the line, including the slack, must be kept in front of the body, between the handler and the load.
- Tag lines must be held in such a manner that they can be quickly and totally released. They must not be looped around wrists or other parts of the body.
- Tag lines must not be secured or attached in any manner to adjacent structures or equipment. This includes the practice of making a round turn on stanchions or similar structures and surging the line to control the load.
- A push/pull pole shall be used to retrieve the suspended tag line.

All cargo will be pre-slung at the onshore supply base. Wire rope slings will be the norm, chain slings will not be accepted. Man-made fibre slings will NOT be accepted as the primary lifting slings for cargo. Their use is restricted to local use on the worksite. Cargo both on and offshore will be subject to inspection checks. The QA/QC Flowchart in Appendix 7 describes this process and provides acceptance and pre- use inspection checklists for QA/QC.

Non-conformance with this Procedure shall result in cargo not being delivered until the necessary remedial actions have been carried out in conjunction with the specific company. In the event of a non-conformance being recognized, the Non-Conformance Report (NCR) shall be completed and communicated with the relevant parties.

When dealing with cargo carrying units (CCUs), guidance on the design of cargo containers and baskets can be found in the Step Change in Safety – Design and Handling of Cargo Baskets, which sets the standard for new / future equipment.

Although most containers have four pad eyes, there are certain items of handling equipment that are better suited to a single pad eye / single point lift. Refer to clause 3.8 of QatarEnergy Standard for Lifting Equipment and Operations (QP-PAI-STD-005)

Note: The unloading of palletized loads from containers should be by forklift truck or by using a fork attachment suspended from the crane.

Note: SLINGS MUST NEVER BE HOOKED DIRECTLY ON TO FORKS, a special jib attachment must be used.

Awkward cargo such as plastic drums can be moved about the worksite using certified cargo nets, preferably the wire rope type.

4.1.1 Textile Sling (Round Sling/Webbing Sling)

- Textile slings shall not have any colour code painted or be marked with ink directly on the sling material. A label or a tag must be attached to indicate the current colour code.
- Textile slings can only be used for a maximum of four (4) years from the initial certification.
- Textile slings shall not be used for offshore transportation.

4.1.2 Acceptance of Crane

Function tests of the crane shall be carried out without load and any defects or adjustments corrected in accordance with the manufacturer’s recommendations on arrival to site.

4.1.3 Daily Crane Checks

Prior to any function testing a visual examination of the crane shall be carried out per the checklist in Appendix 3 Daily Offshore and Mobile Crane Checklist. This and all subsequent actions required for the test must be performed by the crane operator.

4.1.4 Forklift Truck Operations

All Forklift trucks working on QatarEnergy-IS worksites must meet with the minimum requirements specified below.

The banksman controls the initial lifting of the load, laydown of the load and lifts that are out of the line of vision of the Forklift operator.

Forklift trucks will not be left running nor left with the keys inside.

Certified fork attachments must be used when required, e.g. when the forks cannot be used to directly handle the load.

Forklift trucks may only be used for man riding, when supported by risk assessment.

The forklift truck must have the following documentation available for inspection:

1. Current certificate for roadworthiness (if applicable).
2. Current proof load test certificate (i.e. tested within the last 4 years).
3. Proof load test certificates for the mast lifting roller chains.
4. Proof load test certificates for any attachments.
5. Completed and up-to-date daily / weekly maintenance schedules.

Any forklift truck working on an QatarEnergy-IS worksite must also meet with the minimum requirements specified below.

Only FLTs equipped with the following design features shall be used:

1. Automatic audible warning devices that activate when the FLT moves backwards. Similar devices for forward movement can be considered.
2. An orange flashing beacon light that is activated when the ignition switch is moved into the ON position.

3. Operating lights in full working condition at all times.
4. Rear view mirror(s).
5. Speed governors (particularly for all-terrain duty).
6. Hour meters for measuring operating hours and service intervals.
7. An approved two-point seatbelt with appropriate roll over protection cage.
8. Appropriate decals to be pasted on dashboard / rollover cage covering rollover position by operator as prescribed by the manufacturer.
9. A Master Battery Isolation Switch.
10. Signage stating Authorised Operators Only.

When selecting an FLT, consideration needs to be given to:

1. Stability (weight of materials to be handled, outreach required, nature of terrain, etc).
2. The guarding of dangerous parts.
3. Noise generated by the FLT.
4. Operator comfort, protection from the z and falling objects.
5. Air suspension seating or other suitable sprung seating to protect operator from back or similar injury.

Maintenance shall be in accordance with Manufacturers Standards for:

1. Daily checks
2. Weekly checks
3. Prescribed service intervals

Maintenance routines shall cover the condition of brakes, lights, horns, hydraulic systems and tires, and the integrity of the structure and moving parts.

Refer to checklist (Appendix 9) for contracted forklifts arriving on site.

Refer to checklists (Appendices 10/11) for pre-start-up checklists for electric and diesel forklifts respectively.

4.1.5 Man-riding Operations

4.1.5.1 Man-riding winches

All winches used for lifting people shall be certified by an QatarEnergy-IS approved 3rd party with PTS stamp and shall be clearly marked SUITABLE FOR LIFTING PEOPLE or SUITABLE FOR MAN- RIDING. Any winch that is not marked should not be used for lifting people.

The following criteria should be considered for all winches:

The winch-operating lever should automatically return to neutral when released. Automatic brakes should be fitted so that they apply whenever the operating lever is

Returned to neutral or if there is loss of power to the drive or control system.

The following criteria should also apply when winches are used for lifting people:

- A second independent brake should be provided for use if the automatic brake fails. This brake should be manually operated unless it is completely independent of the automatic braking system.
- The winch should have adequate capacity to handle the line load itemized below:
    - The nominal weight of passengers
    - The tare weight of the carriers
    - The rope weight and friction effects

- The sum of these weights should include an adequate safety factor together with a dynamic factor approved by a competent person.
- The winch should be capable of lowering the carrier in a controlled manner in an emergency.
- The brake holding capacity should be less than that generated by the minimum braking load of the rope and greater than that identified by the maximum line forces identified above.

When winches are used for lifting people, the written operating Procedure should include:

- Authority for use
- Means of communication between the winch operator and passenger(s)
- Safety arrangements
- Limiting conditions of use

The operating Procedures should include an assessment made by a competent person to ensure that any uncontrolled ascent of the carrier is prevented under all circumstances. This assessment should include calculations to ensure that the weight of the rope on the winch side of the sheave system is never greater than the minimum weight on the carrying side of the sheave system. This should include an adequate factor of safety.

The following criteria also apply:

- Rope entanglement and undue wear should be prevented.
- The rope should remain captive at all times around the sheaves and at the winch drum.
- The drum should be guarded to protect the operator if the rope should fail (the guard should not prevent the operator from viewing the spooling of the rope). Clutches or other methods of disengaging the drive system are prohibited. Where winches are to be used to lift people, a supervisor and / or winch operator should be able to see them at all times. The supervisor should remain in direct communication with the winch operator throughout the operation

4.1.5.2 Personal Carriers / Work Baskets / Platforms

The design requirements for man-riding baskets are as follows:

- A minimum height from the floor to the underside of the top handrail to prevent persons toppling over. (Minimum height = 910 mm but recommended height of 1150 mm preferred).
- A maximum vertical distance between the handrail / intermediate rails to prevent persons falling through. (Max = 500mm).
- A minimum floor area for each person working in the basket. (Minimum area per person recommended = 600mm X 600mm).
- A stipulation that toe boards must be fitted all around the flooring if the basket is open sided. (Minimum depth 150mm).
- Internal handrails to prevent hands / fingers being trapped if the basket swings against an obstruction.
- Doors (if fitted) must open inwards and have a locking mechanism to prevent inadvertent opening.
- The base of the basket should have rubber buffers to prevent jarring of the spine when being landed.

- It should have internal anchor points for safety harnesses.
- Slip resistant floor and adequate drainage.
- Items carried on the floor of the carrier are effectively secured.
- The primary load line and secondary back up line has the requisite safety factor. Where secondary or back-up lines are used, a risk assessment should ensure that these do not increase the risk. Load lines should be selected to ensure that the carrier does not spin or turn when lifted. All connections between the carrier and the lifting appliance should be unable to self-release under any circumstances.

Note: The basket and accessories must also be inspected immediately before each use by a competent person and the personnel being transferred.

QatarEnergy-IS personnel transfer basket will be the Rigid Billy Pugh model X904.

4.2 PERSONNEL TRANSFERS USING PERSONNEL TRANSFER BASKET / DEVICE AND CRANE

Under certain conditions, a personnel basket / device can be used:

- To transfer personnel via crane from the deck of a marine vessel to a designated landing area on the facility.
- To transfer personnel via crane from a designated landing area on the facility to the deck of a marine vessel.

Personnel transfers shall be discussed between and approved by the appropriate personnel at the facility (OIS, ODS, etc) and the marine vessel captain.

In determining the suitability of transfer by basket, the following factors shall be taken into account:

- Weight, health condition, etc. of the persons to be transferred
- Suitability of the vessel and crane for the proposed transfer
- The personnel basket / device and its equipment is certificated and tested, the
- SWL is clearly marked on it and there is no visible damage
- All cranes are fitted with an approved safety hook
- Limitations as to visibility and sea state
- Necessity for performing a transfer during the hours of darkness
- Availability of the standby vessel or fast rescue craft
- Wind speed and vessel movement limitations on crane operations
- Competence of personnel transfer supervisors and others involved
- All equipment to be employed has been tested and/or inspected
- A suitable clear deck space exists with no masts, aerials or other obstructions in or around the landing area.

The lift shall be carried out over the sea wherever practicable

Personnel transfer supervisors shall control the personnel transfer on the facility and on the marine vessel.

Personnel transfer supervisors shall be in radio contact with the crane operator, with the marine vessel captain and with each other.

A maximum of four (4) persons and their personal luggage may be transferred at one time via personnel basket / device.

Personal luggage shall be stored inside the personnel basket / device and shall not be carried by hand.

Personnel transfer Supervisors shall:

- Ask all personnel about to be transported by personnel basket / device, if they are familiar with the correct Procedure
- Ensure that the transfer Procedure is understood and all personnel are agreeable to the transfer
- Ensure the safety equipment provided is correctly used
- Ensure that personnel are experienced in the form of transfer or are accompanied by an experienced person/
    - Shall provide instruction in the proper Procedure, if required.

Personnel baskets / devices shall have four (4) approved life jackets stored inside the basket / device, ready for donning and use by personnel being transferred.

All personnel baskets / devices shall be equipped with two tag lines (manila rope, 4 –

6 meters in length, unknotted) to assist the personnel transfer Supervisor in guiding the personnel basket / device safely from and to the landing area.

When a personnel basket is used for the transfer, personnel shall don their life jackets, stand on the rim of the circular base and shall grasp the basket netting at shoulder height with both hands, prior to the lift.

Personnel basket / device transfers shall not be undertaken during conditions of poor visibility (inadequate lighting, fog, heavy rain, etc.)

4.2.1 Personnel Transfer from / to Marine Vessels

Prior to approaching the facility, the marine vessel captain shall discuss the personnel basket / device transfer via radio with the appropriate personnel on the facility.

The appropriate personnel on the facility and the marine vessel captain shall ensure that only trained, designated personnel are authorized to supervise the personnel transfer.

Where a transfer of personnel via personnel basket / device is anticipated, an area of the marine vessel deck shall be kept cleared to ensure the personnel basket / device has adequate clearance for safe pick-up and set-down via the crane.

The personnel transfer shall proceed only:

- When the marine vessel is properly positioned
- When radio contact has been tested
- When all participants (crane operator, personnel transfer supervisors and marine vessel captain) are prepared and agree that the transfer can be conducted in a safe manner.

After the personnel basket / device is raised to the minimum height to clear obstructions, the crane boom shall be slewed away from the facility or marine vessel, to ensure that the personnel basket / device is raised and lowered over water and not over the facility or marine vessel.

While being raised or lowered, the personnel basket shall be far enough away from obstructions such that it cannot swing and strike any obstruction.

4.2.2 Personnel Transfers during Darkness

- Transfer operations during darkness shall be avoided whenever possible. However, a transfer during the hours of darkness may be permitted, provided:
  - Transfer areas are adequately illuminated
  - Operations are conducted under the personal direction of the appropriate Person in Charge (e.g. OIS, ODS) and the Vessel Master

4.2.3 Wire Rope Change-out Period

Wire rope will be changed out at the periods given in the table below. The life cycle may be extended if a Monthly Crane Wire Rope Inspection Checklist is being conducted and wire rope condition is monitored within the checklist. The scope of the written scheme extends to all lifting operations and lifting equipment in use on IS's worksites. Further details of specific visual inspection and thorough examination scopes are given in IS-HSE-PRC-027A Lifting Operations Guidance Document.

QatarEnergy-IS Wire Rope Replacement Frequencies		
Description	**Frequency**	**Remarks**
Crane Pennant Wire	4 years	
Boom Wire	1 year	
Main Line	18 months	May be extended to 2 years for cranes with an auxiliary hoist
Whip Line	18 months	
Rig Floor Utility Winches	1 year	
Man Riding Winches	1 year	
BOP Set Back Winches	1 year	
Lifeboat / Life raft fall wires	4 years	*May be extended to 5 years provided thorough wire rope inspection & lubrication programs are in effect and results are documented.
Anchor Winch	5 years	
Crane Hook Blocks	4 years	Shall be overhauled, inspected and tested.

4.2.4 Visual Inspections

Visual inspections are inherently not well suited for the detection of internal wire rope deterioration. Therefore, they have limited value as a sole means of wire rope inspection.

Each monthly visual inspection conducted by a competent person shall include as a minimum

- Date of examination.

- Broken wires. Record sufficient details to identify compliance with discard
- criteria:-
- The number of wire breaks per lay length that occur in a strand.
- The location of the lay length with the greatest number of wire breaks.
- The total number of wire breaks in the wire rope.
- Details of broken wires near to any termination point.
- The rate of increase of broken wires.
- Wear.
- Corrosion.
- Change in wire rope diameter.
- Change in lay length.
- Distortion.
- Identification and signature of examiner.

The monthly findings shall be entered into Appendix 2 Monthly Crane Wire Rope Inspection Checklist. The checklist shall be retained by QatarEnergy-IS and the third-party examiner until the wire rope is retired from service. Pressure lubrication of wire ropes is recommended

4.2.5 Pad Eyes

Pad eyes shall be proof load tested on initial installation before being put into use, after reinstallation at the site, and at the discretion of the surveyor. Following proof load test, MPI shall be carried out.

A visual and thorough examination shall be carried out at six monthly intervals. Following thorough and visual examination and if defects are evident, the third party

certifying authority may instruct to carry out further tests and NDT at his discretion to

assess the integrity of the pad eyes.

There are three different pad eyes categorization, namely:

1. Permanent Pad Eyes - Pad eyes permanently attached to any lifting appliances with design calculation. Each pad eye shall be tested to two times SWL
2. Fabricated Pad Eyes - Pad eyes attached to any lifting equipment without proper design calculation or unknown pad eyes (missing identification). Each pad eye shall be tested to three times SWL
3. Temporary Pad Eyes - Pad eyes intended for lifting of offshore jackets, modules, etc. Requirements for Proof Load Test may be deviated without a deviation from being raised - provided the concerned Department or contractor can demonstrate that the pad eyes have been designed, installed and inspected in accordance with the requirements of the QatarEnergy-IS Structural Engineer.

4.2.6 Organizing of Statutory Examinations

These personnel will advise all other relevant personnel of the impending examinations to allow them to prepare and make available all lifting equipment and accessories. The Senior Worksite Supervisor will then mobilize the specialist personnel to perform the inspections and will help co-ordinate activities by providing administration services for any remedial work required on the lifting equipment / accessories. They are also responsible for maintenance of all lifting equipment test and examination records. Where equipment has no specified inspection date, then the competent person for lifting operations shall request, prior to use, that a thorough examination is carried out by a suitably qualified person.

4.2.7 Personnel Conducting Examinations

Statutory examinations on fixed & loose equipment will be carried out by third party specialist lifting companies who have been approved by QatarEnergy-IS QA/QC Department (Appendix 7).

Any reference in this Document to the inspection, verification, training and certification of any lifting equipment or lifting equipment personnel, shall mean the issue of such Documents by one of the QatarEnergy-IS approved bodies listed above.

Accessories or portable lifting equipment in rigging lofts will be changed out every 6 months.

4.2.8 Methods of Examination and Examination Procedures

The methods of thorough examination / inspection and testing for the purpose of this written scheme are as follows:

a) Visual Inspection
b) NDT Inspections
c) Load Testing

4.2.9 Color Coding of Equipment

QatarEnergy-IS operates a system whereby ALL Lifting Equipment is color coded with a unique color, at six (6) monthly intervals after inspection where applicable. This color is confirmed and validated by memorandum and prominent display boards. Equipment not coded in accordance with the required color shall not be utilized in QatarEnergy operational areas.

Contractors operating within QatarEnergy-IS operational areas shall comply with these color code requirements.

QatarEnergy-IS operates a 30 days' grace period for transit equipment during color code transition dates.

Condemned equipment will be color coded RED until such times as it can be repaired, from the worksite.

Note: Color coding is only an indicator that the equipment certification is current. **IT DOES NOT MEAN THAT THE EQUIPMENT IS SAFE TO USE**. It may have been damaged since the last formal inspection, therefore a safety inspection must be carried out immediately prior to each use.

4.2.10 Condemned Gear – Quarantine Procedure

Any lifting equipment found to be defective during an inspection (or at any time during normal use) must be removed from service immediately until such times as the defect can either be remedied or the item can be removed from the worksite. If practicable, it should be destroyed on discovery. Notification of defective equipment must be given to the relevant Supervisor who will arrange to repair or return the defective equipment to the onshore rigging loft for repair or scrap.

All worksite should have a controlled quarantine area where condemned equipment is safely stored to prevent inadvertent use.

4.2.11 Actioning of defects list

A work order is raised to repair defective fixed equipment. Defective portable equipment is put in quarantine and then back-loaded for repair.

4.2.12 Information Required for Database

Statutory examination report forms and / or defect lists will be supplied by the 3rd party independent lifting equipment Inspection Company. When completed, the reports must contain enough information to fully describe the item, its condition / suitability for further service, any defects and any recommendations.

The defects list must contain the following information:

- Location
- ID Number
- Description
- SWL / WLL
- Defect / reason for taking out of service
- Date and signature of person conducting the examination.

5.0 REFERENCES TO OTHER DOCUMENTS

Document Title	Document Number	Internal/External Document	Document Referencing
Lifting Operations Guidance	IS-HSE-PRC-027A	Internal	Downward
Integrated Safe System of Work (ISSOW)	IS-FOP-PRC-100	Internal	Sideward
NSL International Rigging and Lifting Handbook	-	External	N/A
NSL– Safe Cargo Handling Good and Bad Practice pocketbook	-	External	N/A
Specification for Offshore Pedestal Mounted Cranes	API – RP 2C	External	N/A
Operation and Maintenance of Offshore Cranes	API – RP 2D	External	N/A
Recommended Practice on Application	API – RP 9B	External	N/A
Corporate Procedure for Lifting Equipment and Operations	CORP-ENG-PRC-038	Internal	Upwards
Corporate Standard for Lifting Equipment and Operations	QP-PAI-STD-005	Internal	Upwards

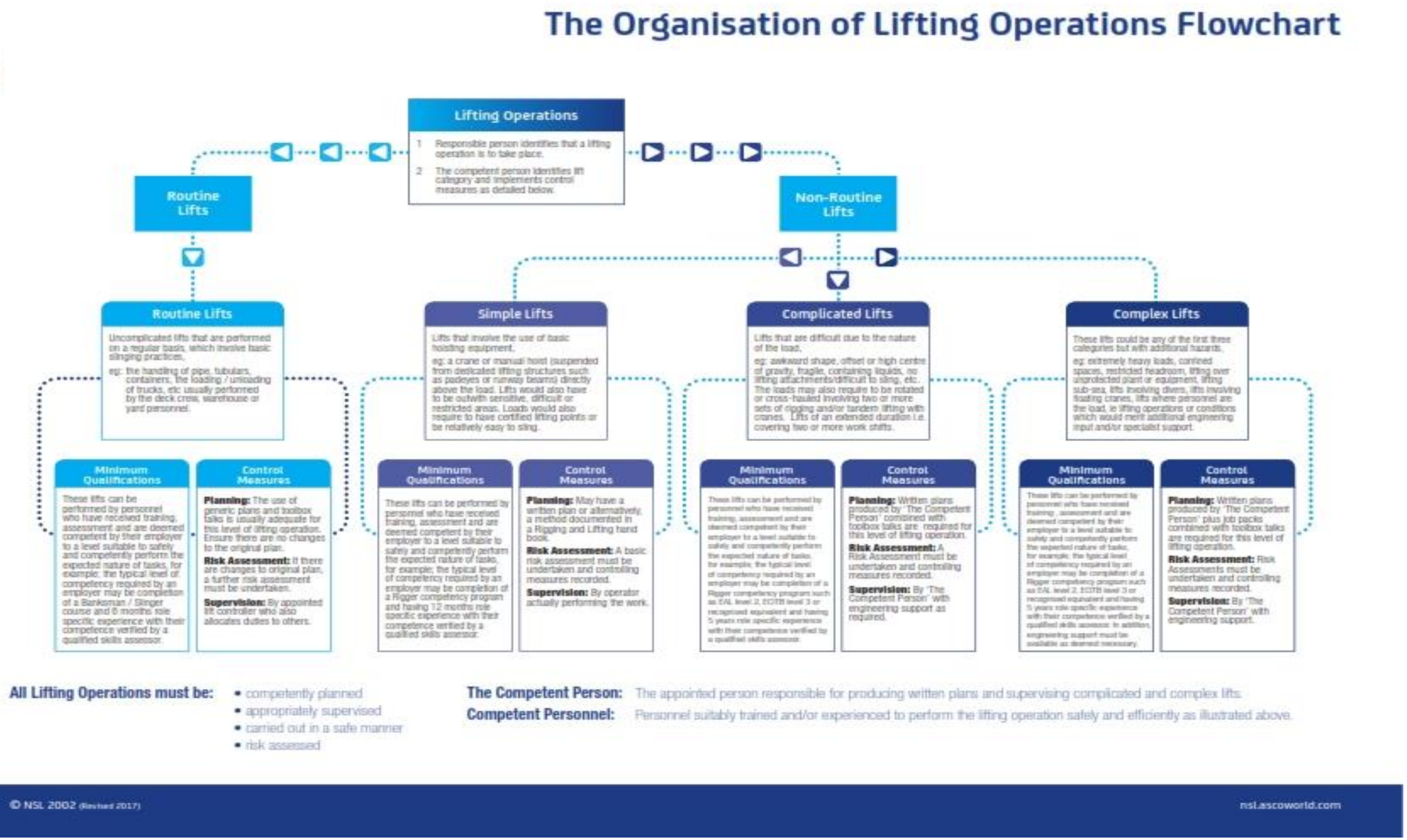
6.0 DEFINITIONS

Term	Definition
ERRV (Emergency Response Rescue Vessel)	The ERRV is a purpose-built emergency vessel equipped to respond to emergencies at offshore locations. The ERRV is normally stationed on the Marina on its davit. It can also be stationed at Halul Island depending on the situation. The ERRV can assist in emergencies and in the recovery of personnel in man-overboard incidents. It also has a highly specialized trained crew.
Fast Rescue Craft (FRC)	A purpose-built motor launch, mounted on a standby vessel and used to assist in the safe recovery of personnel who have fallen from work platforms or marine vessels into the sea.
Helideck	A purpose-built landing area for helicopter operations on drilling rigs, wellhead jackets, barges and other offshore installations.
Lifting Accessories	Components for attaching loads(s) to lifting appliances or machinery e.g. slings and shackles, including: • Wire rope / chain / webbing slings • Wire rope • Shackles • Swivels • Rigging screws • Vertical / horizontal plate clamps • Lifting beams • Sheave blocks • Inertia reels and safety harnesses • Collar eyebolts • Man-riding baskets/passenger cradles & belts • Rope access equipment • Safety / C / swivel / pipe hooks • Crane hook blocks
Lifting Appliances	Appliances or machinery capable of raising, suspending or lowering a load, including any attachments for anchoring, fixing or supporting the appliance, e.g.: • Barrel lifters • Mobile & pedestal cranes / overhead gantry cranes • Air / electric / hydraulic / manual chain (chain block) / wire rope (Tirfors) / lever hoists (Pullifts) • Combined hoists and trolleys • Runway beams

Term	Definition
	• Welded / bolted pad eyes • Air / hydraulic / manual winches • Scissor lifts • Forklift truck & forklift truck lifting attachments • Mobile elevating work platform • Beam trolleys / clamps
Marine Vessel (Work Boat, Supply Boat, Pipe-lay Barge, Derrick Barge, Lift Boat)	Ocean-going self-propelled boats used to transport materials and personnel to / from offshore facilities and to conduct specific work activities.
Mobile Offshore Drilling Unit (MODU)	A mobile offshore drilling unit that is moved on water between work locations.
Mobile Offshore Unit (MOU)	A well servicing unit, accommodation unit, pipe-lay barge, etc. having a marine hull, and moved on water between work locations.
Other Personnel Transfer Devices	Devices of solid construction (base plate with sidewalls) designed and certified for the transfer of personnel from one level to another via crane
Personnel Transfer Basket	A device used to transfer personnel from one level to another via crane. Consists of a solid outside ring around a soft central platform 6' in diameter with netting fastened to it above the base. The personnel transfer basket attaches to the crane by a cable fastened to a lifting ring at the top of the netting.
Rig or Self- Elevating Barge	A work unit which is placed alongside a wellhead structure and is required to maintain its work deck several meters above the water level such that a crane and a personnel transfer basket may be required to bring personnel on board, from a marine vessel.
Wellhead Jacket (WHJ)	An installation containing wellhead equipment, associated piping, valves and control equipment, supported on a steel riser structure, and which contains a marine vessel boat landing deck.

APPENDIX

APPENDIX 1 - LIFTING OPERATIONS FLOWCHART



APPENDIX 2 - MONTHLY CRANE WIRE ROPE INSPECTION CHECKLIST

Monthly Crane & Wire Rope Inspection Checklist

Inspection results; *(Initial each box as check completed)*

Pedestal Cranes			
Crane Operator: **Date of check:**	**Location**	**Location**	**Location**
Wire rope shall be checked for kinking, crushing, broken strands and Internal & External corrosion.			
Rope should be checked for any reduction in diameter due to loss of internal support, corrosion or wear.			
Wire rope should be adequately lubricated at all times.			
Rope end terminations are to be in good order, secure and free from corrosion.			
Hoist hook shall be visually checked for damage or cracking. Hook should freely rotate and locking latch should be in good order. Hook throat opening should be visually checked for indication of			
Rope drums and guides to be checked for wear and correct reeving of rope.			
Sheaves grooves to be gauged/checked for proper size/wear and logged accordingly.			
Sheaves to be greased, freely rotating and checked for signs of wear or damage.			
Using a Vernier Caliper Gauge, measure rope diameter (always-select the part of the rope that travels the most through sheaves – each crane will be specific according to the operational conditions and reeving). Log findings in Wire Rope inspection log.			

MONTHLY CRANE WIRE ROPE INTERNAL & EXTERNAL INSPECTION LOG

<table>
<tr><td colspan="4">RIG NAME:</td><td colspan="4">CRANE SERIAL NUMBER</td></tr>
<tr><td colspan="4">CRANE MAKE/MODELJTYPE:</td><td colspan="4">ROPE APPLICATION (MAIN/AUX/BOOM):</td></tr>
<tr><td colspan="4">CONSTRUCTION:</td><td colspan="4">DATE FITTED:</td></tr>
<tr><td colspan="4">DIRECTION OF ROPE LAY: RH I LH [A]</td><td colspan="4">DATE DISCARDED:</td></tr>
<tr><td colspan="4">TYPE OF LAY: ORDINARY / LANGS [A]</td><td colspan="4">REASONS FOR DISCARD:</td></tr>
<tr><td colspan="4">NOMINAL DIAMETER:</td><td colspan="4">MINIMUM BREAKING LOAD:</td></tr>
<tr><td colspan="4">TENSILE GRADE:</td><td colspan="4">WORKING LOAD:</td></tr>
<tr><td colspan="4">FINISH: BRIGHT / GALVANIZED [A]</td><td colspan="4">SHEAVES MATERIAL:</td></tr>
<tr><td colspan="4">TYPE OF CORE (IWRC, WSC, FC, PC, FFC):</td><td colspan="4">DIAMETER MEASURED:</td></tr>
<tr><td colspan="4">NUMBER OF SHEAVES IN SYSTEM:</td><td colspan="4">UNDER A LOAD OF:</td></tr>
<tr><td colspan="4">LENGTH OF ROPE:</td><td colspan="4">WINCH DRUM TYPE (grooved/non-grooved):</td></tr>
<tr><td colspan="4">TYPE OF TERMINATION:</td><td colspan="4">TYPE OF LUBRICATION (manual/pressurized):</td></tr>
<tr><td>VISIBLE BROKEN WIRES</td><td>ABRASION OF OUTER WIRES</td><td>CORROSION</td><td>REDUCTION OF ROPE DIAMETER</td><td>POSITIONS MEASURED</td><td>NDT (YES/ NO)</td><td>OVERALL ASSESSMENT</td><td>DAMAGE AND DEFORMATIONS</td></tr>
<tr><td>NUMBER IN LENGTH OF 6d</td><td>DEGREE OF DETERIORATION [B]</td><td>DEGREE OF DETERIORATOIN [B]</td><td>%</td><td></td><td></td><td>DEGREE OF DETERIORATION [B]</td><td>NATURE</td></tr>
<tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr>
<tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr>
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<tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr>
<tr><td colspan="3">INSPECTOR NAME:</td><td colspan="5">DATE:</td></tr>
<tr><td colspan="3">ROPE SUPPLIER:</td><td colspan="5">NUMBER OF WORKING HOURS:</td></tr>
</table>

APPENDIX 3 - DAILY OFFSHORE & MOBILE CRANE CHECKLIST

Inspected By:	
Signature:	
Date:	
Crane Make/Model:	

		Y	N	N/A	Initial
Initial Checks					
Access	Clear				
Ladders, Hatches And Walkways	Clean, secure and unobstructed				
Gratings And Deck-Plates	Clean and secure				
Machinery House	Clean and tidy				
Operator Cabin	Clean and tidy				
Windscreen	Clean				
Windscreen Wipers	Check operation and effectiveness				
Pre-Start Checks					
Boom Chords And Lacings	Check for damage				
Pendant And Hook Blocks	Check condition				
Wire Ropes And Terminations	Check condition and security				
Machinery Guards	Check in place and secure				
Oil, Fuel And Coolant	Check levels				
Pre- Operation Checks					
Air/oil pressure	In safe operating range				
All controls, including brakes	Check for safe operation				
Boom radius cut-out/alarms	Check for safe operation				
Hook block cut out alarm	Check for safe operation				

Brake/clutch linings	Check for contamination				
Safe Load Indicator					
Audible/Visual Alarms	Check operation				
Load Radius Chart	Displayed in cab				
Weight Displayed	Check accuracy				
Communication					

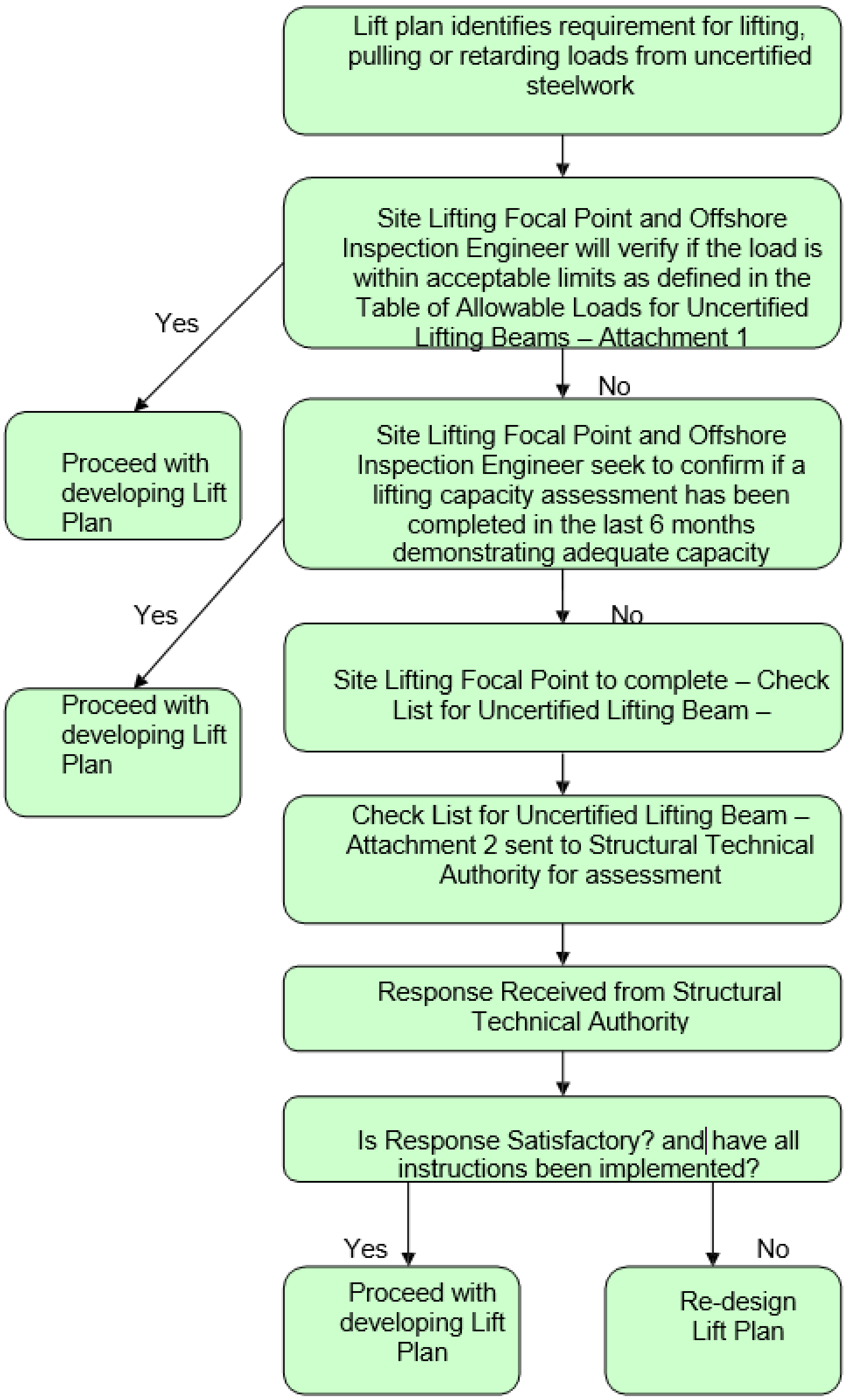
VHF/UHF Radios	Test transmission				
Rigger	Discuss hand signal				
Crane Signal Charts	Displayed cab				

Other Activities/Conditions					
Within Crane Radius	Confirm no conflict – (i.e. power-lines)				
General Activities	Radio silence, helicopter operations, wire-line activities, other cranes				
Site Conditions	Out rigger to be checked for support area and additional support to be provided where necessary				
Wind/Sea State/ Visibility	Within limits				
Safety Equipment/Features					
Fire Extinguisher	Available and in-service				
Floodlights and Aviation Lights	Check working				
SCBA Set	Stowed in cab				
Life Jacket/Vest	Stowed in cab				

ALL DAMAGE, PROBLEMS OR MALFUNCTIONS ARE TO BE IMMEDIATELY REPORTED TO YOUR SUPERVISOR AND LOGGED INTO THE DAILY MAINTENANCE LOGBOOK

APPENDIX 4 - ASSESSMENT OF NON-CERTIFIED LIFTING POINTS

PROCESS FOR LIFTING OPERATIONS FROM UNCERTIFIED STEELWORK



1. INTRODUCTION

The guideline gives clarification to the process for determining the load capacity in existing structural steelwork to act as a lifting point for lifting operations. This guideline shall be read in conjunction with QatarEnergy-IS Lifting Procedures.

2. PURPOSE

The purpose of this guideline is to determine the process for sites to establish load capacities for existing structural steelwork to be used in lifting operations. It is critical to assess pre-existing loads in structural steel work before the addition of further loading from lifting operations to maintain the steelwork integrity within design code and minimise the risk of damage or dropped objects

3. SCOPE

It applies to lifting operations carried out at all QatarEnergy-IS operated sites.

4. PROCESS

Once an anchor point is identified as uncertified steelwork, it is the responsibility of the Site Lifting Focal Point and the Offshore Inspection Engineer to verify that the intended load is within acceptable limits as defined by the Structural Technical Authority table of allowable loads. (Attachment 1)

If the load is within the defined limits then the lifting plan may be further developed.

If it is out with allowable limits then the Site Lifting Focal Point will check to see if a load assessment has been completed for that anchor point in the last 6 months.

If an assessment has been found and permits the desired load then the lifting plan may be further developed.

If there is no pre-existing load assessment then the Site Lifting Focal Point shall complete a check list for uncertified lifting and submit it to the Structural Technical Authority for assessment. (Attachment 2)

Upon receipt of a response from the Structural Technical Authority it shall be reviewed to determine if the response is satisfactory, and the lifting plan will be developed to completion. If there any conditions, restrictions or mitigation required from the Technical Authority then they shall be implemented prior to the commencement of any lifting operations

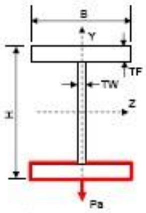
If the response does not permit the proposed loading then an alternative arrangement and lifting plan will require to be developed.

ATTACHMENT 1 - TABLE OF ALLOWABLE LOADS FOR UNCERTIFIED LIFTING BEAMS

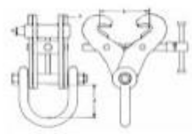
ATKINS

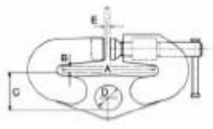
TABLE OF ALLOWABLE LOADS FOR UNCERTIFIED LIFTING BEAMS

SECTION NUMBER	LIMITING SECTION SIZES				ALLOWABLE VERTICAL LIFT LOAD	MAXIMUM ALLOWABLE SPAN
	H	B	TW	TF	Pa	L
	mm	mm	mm	mm		(mm)
1	100	55	4.1	5.7	170 kg	1500
2	160	82	4.7	7.4	290 kg	2500
3	200	100	5.2	8.5	380 kg	3000
4	300	150	6.1	9.2	400 kg	5000
5	330	160	6.9	9.7	0.5 Te	5000
6	400	175	7.6	10.9	0.6 Te	6000
7	450	180	7.6	10.9	0.6 Te	6000
8	500	200	9.6	13.2	0.9 Te	7000
9	550	210	10.6	13.2	0.9 Te	7000
10	600	220	10.6	14.8	1.2 Te	7000
11	750	263	11.5	17	1.5 Te	9000
12	770	300	14	18	1.7 Te	10000



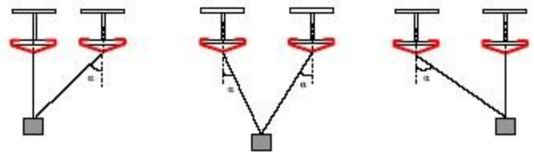
Lifting Beam Details


Clamp Type A


Clamp Type B

NOTE ON NON-VERTICAL LIFT

1) Lifting capacity for non-vertical lifts is half of the allowable lift load.
2) The maximum lifting angle from vertical for Clamp Type A is 15^0 and for Clamp Type B is 30^0.
3) For two point loading the full allowable vertical lift load may be lifted provided that the angle α is within the allowable limit for both lifting cables at all times during the lifting operation (see sketch below).
4) Both ends of the lifting beam must be fully restrained against torsional rotation.
5) If any of the above conditions are not satisfied, then the CSE must be consulted.


Two point loading arrangements ($\alpha \le 15^0$ for Clamp type A; $\alpha \le 30^0$ for Clamp type B)

GENERAL NOTES ON USING THIS TABLE

1) Load must be applied symmetrically to the bottom flange of the lifting beam or supported by a sling around beam.
2) No other loads should be applied to the lifting beam other than minor walkway loads.
3) No pipes greater than 152mm/6" diameter should be supported by the lifting beam.
4) Supporting beams at both ends of lifting beam must satisfy the same requirements as for the lifting beam, and may need to be independently assessed.
5) Lifting beam must be supported at both ends, and must not be part of a cantilever system (e.g. cantilevered walkway).
6) End welds must be continuous around the flanges and web, and must be full penetration or have a leg length at least equal to the web thickness.
7) Lifting beam must be a rolled section, i.e. this guidance is NOT applicable to fabricated or welded beams.
8) An inspection of the beam and end connections must be carried out prior to lift.
9) If any of the above conditions are not satisfied, then the CSE must be consulted.

ATTACHMENT 2 - CHECK LIST FOR UNCERTIFIED LIFTING BEAM

CHECK LIST FOR UNCERTIFIED LIFTING BEAM

Platform	
Location	
Item to be listed	
Weight of the lift	

Lifting Beam Size	
Height	
Width	
Flange thickness	
Web thickness	
Beam section number	
Beam capacity for vertical lift if applicable	
Beam capacity for non-vertical lift if applicable	
Actual beam span	
Maximum allowable beam span from Table	

	Questions	Yes / No
1	Is applied load less than beam capacity ?	
2	Is span less than maximum allowable span ?	
3	Is beam a rolled section ?	
4	Is it a vertical lift ?	
4.1	If no, is note on non-vertical lifts taken into account ?	
5	Has an appropriate beam clamp of Type A or B been used, or is the load supported by a sling around the beam ?	
6	Is beam free from other equipment and laydown loads ?	
7	Is beam free from pipe support loads for pipes >152mm/6" diameter ?	
8	Is beam supported at both ends ?	
9	Are beam end connections free from defects ?	
10	Is beam free from significant corrosion (Cat.A and worse) and physical damage ?	
10.1	If no, is remaining size greater than limiting section size given in Table?	
11	Beam is **Not** part of a cantilever system (answer no if it is part of such a system) ?	

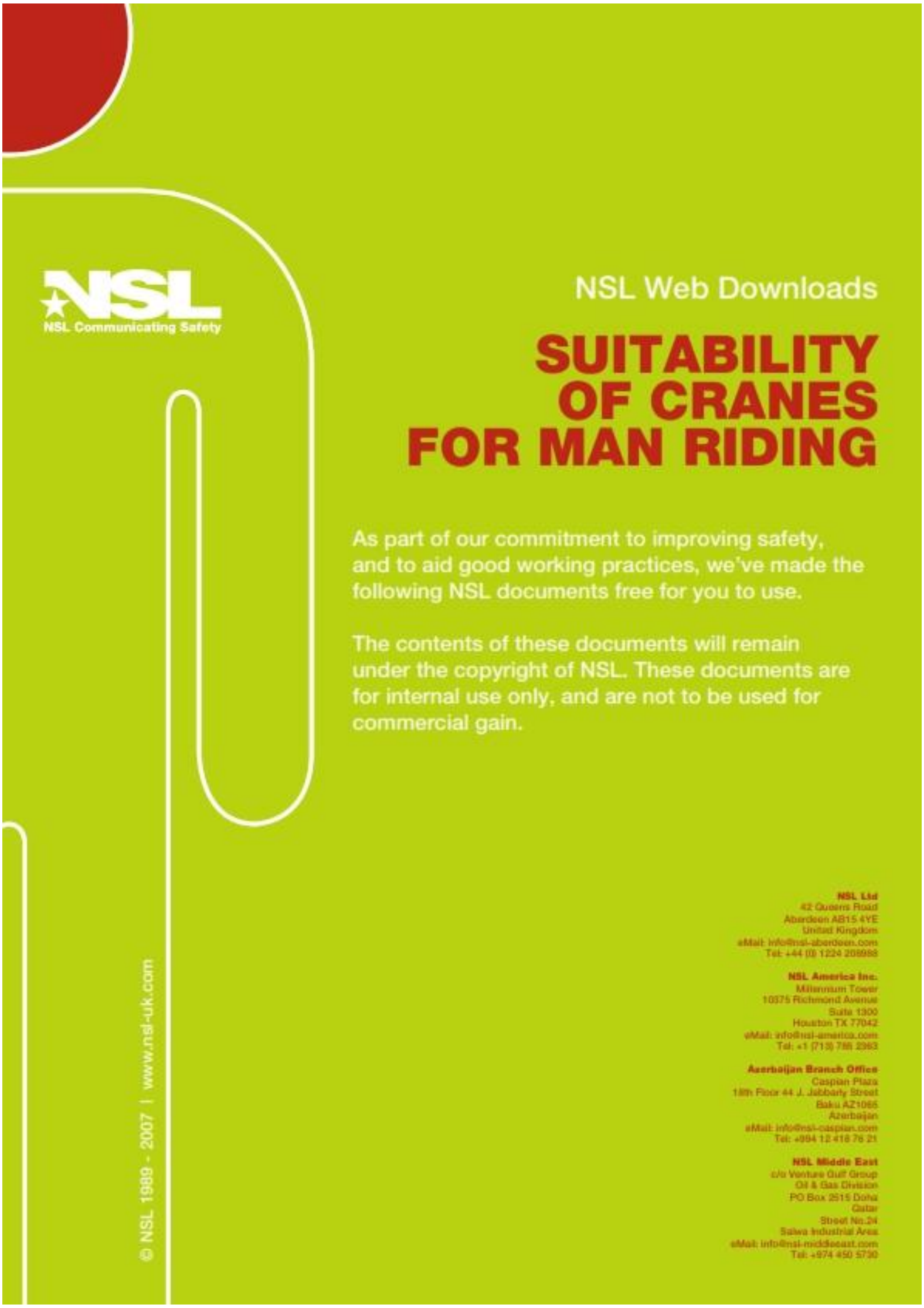
If the answer is **no** to any of the above questions, then consult with the Competent Structural Engineer prior to lift.

Note: If the answers to the questions 4.1 and 10.1 are yes, then the answers to questions 4 and 10 respectively may be ignored.

APPENDIX 5 - CRANE OPERATIONS WEATHER LIMITATIONS

Cranes shall not be utilized when the wind speed is more than 25 knots or exceeds values set out by the OEM of the crane, whichever is lesser. Use of cranes above 25 knots can only be considered on a case-by-case basis, subject to a detailed risk assessment supported in writing by the OEM's certified maximum safe operating wind speed. Under no circumstances are lifting operations permissible in wind speeds in excess of OEM certified maximum safe operating wind speed. All lifting operations above 25 knots require a written acceptance by a QatarEnergy-IS Competent Person.

APPENDIX 6 - SUITABILITY OF CRANES FOR MAN RIDING



NSL
NSL Communicating Safety

NSL Web Downloads

SUITABILITY OF CRANES FOR MAN RIDING

As part of our commitment to improving safety, and to aid good working practices, we've made the following NSL documents free for you to use.

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NSL USER DOCUMENTS
SUITABILITY OF CRANES FOR MAN RIDING

Due to specific verification requirements, the suitability of cranes for man-riding must be assessed for each and every operation. To assist with the assessment, a tick list has been prepared which should be completed by the appropriate competent person appointed by the duty holder under SI 1998 No 2307.

		YES	NO
1	Has it been established that no other viable option of carrying out the work is available?		
2	Are all the necessary certificates for the crane, crane wire ropes, slings and other associated equipment current?		
3	Has the crane and associated equipment been thoroughly inspected by a suitably qualified / competent person within the last 6 months?		
4	Is the crane in good condition, regularly inspected and maintained and are records kept to substantiate this?		
5	Are all the safety features and systems working properly (eg Rated Capacity Indicators (RCIs), overhoist limiters, etc)?		
6	In the event of a complete power failure, will the crane maintain the load in a safe condition (eg do the brakes fail to the applied position)?		
7	Are the brakes applied progressively (eg to avoid shock or snatch loading)?		
8	In the event of a complete power failure, can the load be lowered manually to a position where the personnel can be recovered safely?		
9	In the event of a primary brake or transmission system failure, will the load be prevented from free-falling (eg is there a secondary braking system or does the transmission system have hydraulic retardation to prevent this)?		
10	In the event of the primary brake system failing, can the load be lowered manually to a position where the personnel can be recovered safely?		
11	Is the crane fitted with an emergency stop which is located for immediate operation by the crane operator?		
12	Is the crane so designed that inadvertant freefall is prevented when the drive train is in motion or the hook is loaded?		

Please refer to the reverse side of this document for further instructions

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SUITABILITY OF CRANES FOR MAN RIDING

IF THE ANSWER TO ALL OF THE QUESTIONS ON THE PREVIOUS PAGE IS YES, THE CRANE IS SUITABLE FOR MAN-RIDING AND SHOULD BE MARKED ACCORDINGLY

IF THE ANSWER TO ANY OF THE QUESTIONS OVERLEAF IS NO, THE CRANE IS NOT SUITABLE FOR MAN-RIDING AND MUST BE MARKED ACCORDINGLY

Once it has been established that the crane is suitable for man-riding operations, you must plan the actual operation you are about to perform and carry out a risk assessment in the normal manner.

If the risks have been minimised and deemed acceptable, you can then carry out the lifting operation in accordance with your generic procedures for man-riding operations as long as:

1 The workover basket / transfer basket complies with SI 1998 No 2307.

2 Personnel have been trained / have experience of work / transfer baskets.

3 The crane operator has been trained and assessed as competent for this type of basket / transfer operation.

4 An operator / mechanic capable of operating the crane is available in the event of an emergency.

5 The crane is inspected by the operator prior to the lifting operation.

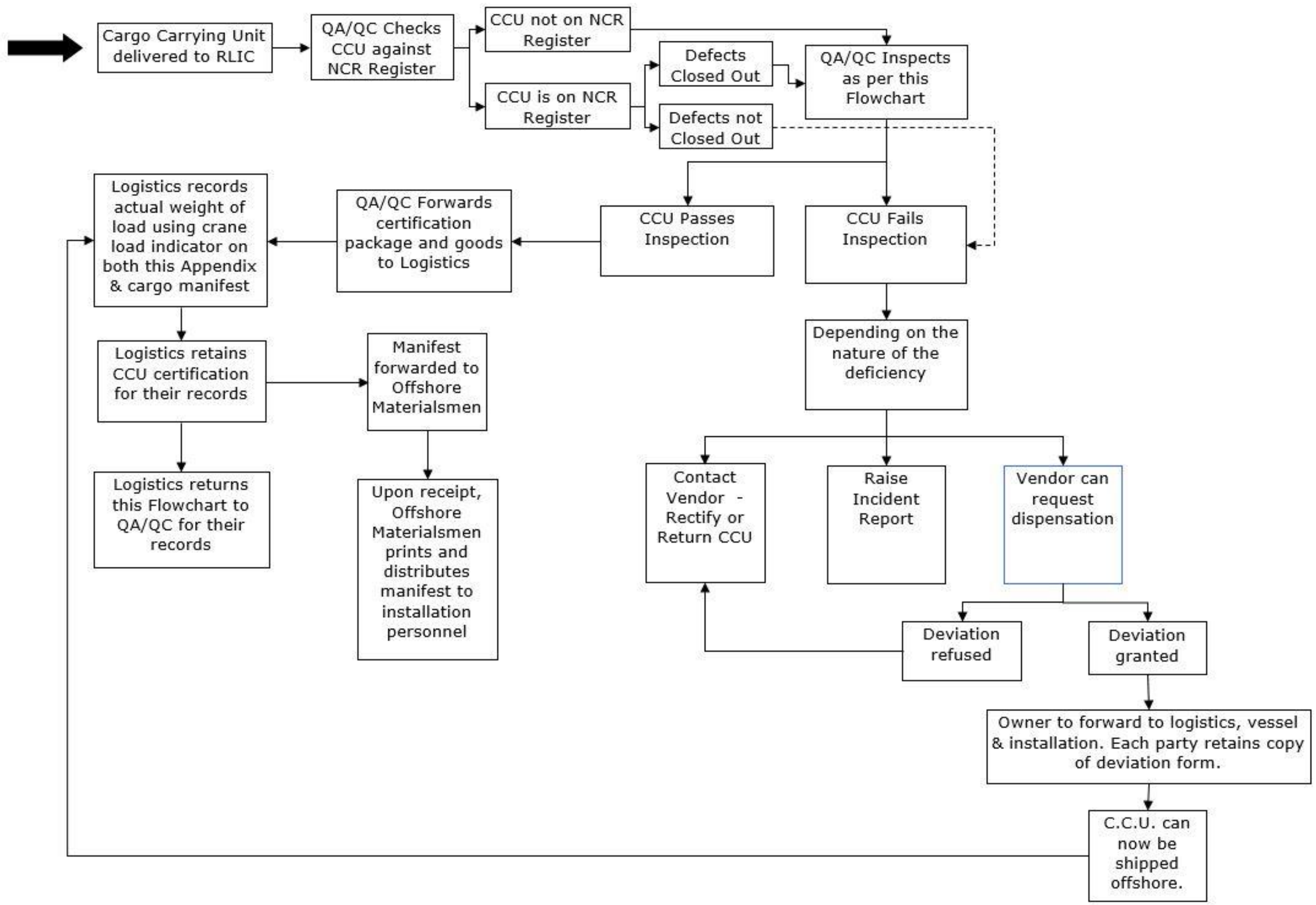
6 A line of communication has been established between the operator and the personnel in the work basket (including a dedicated banksman if required).

7 The environmental conditions have been established by a competent person as being suitable for lifting / transferring personnel by this method.

Note: This document has been viewed and commented on by the UK Health & Safety Executive (Offshore Safety Division). The Executive have no objection in principle to this guidance relating to the suitability of cranes for man-riding purposes being used as good practice within the offshore industry.

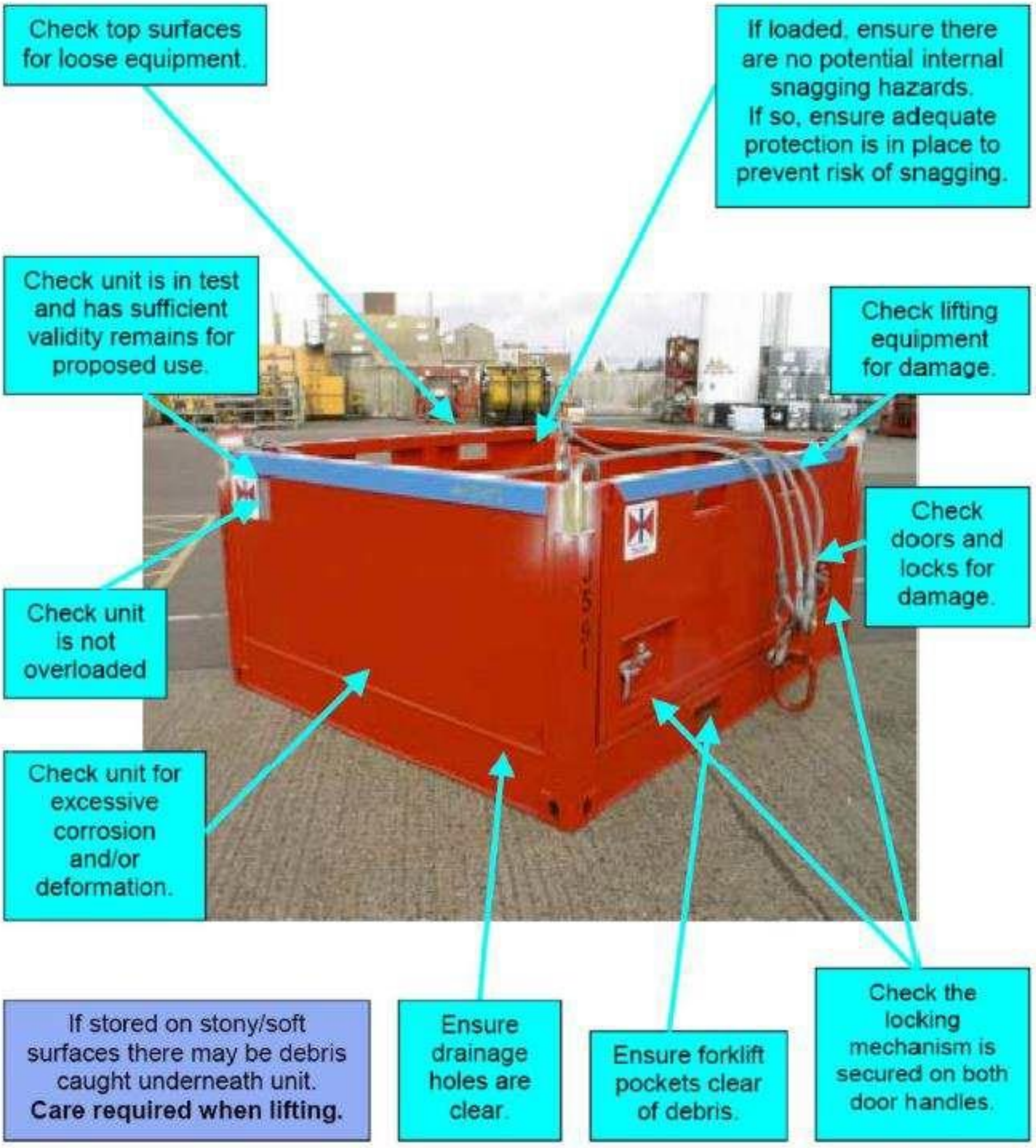
APPENDIX 7 - QAQC FLOWCHART

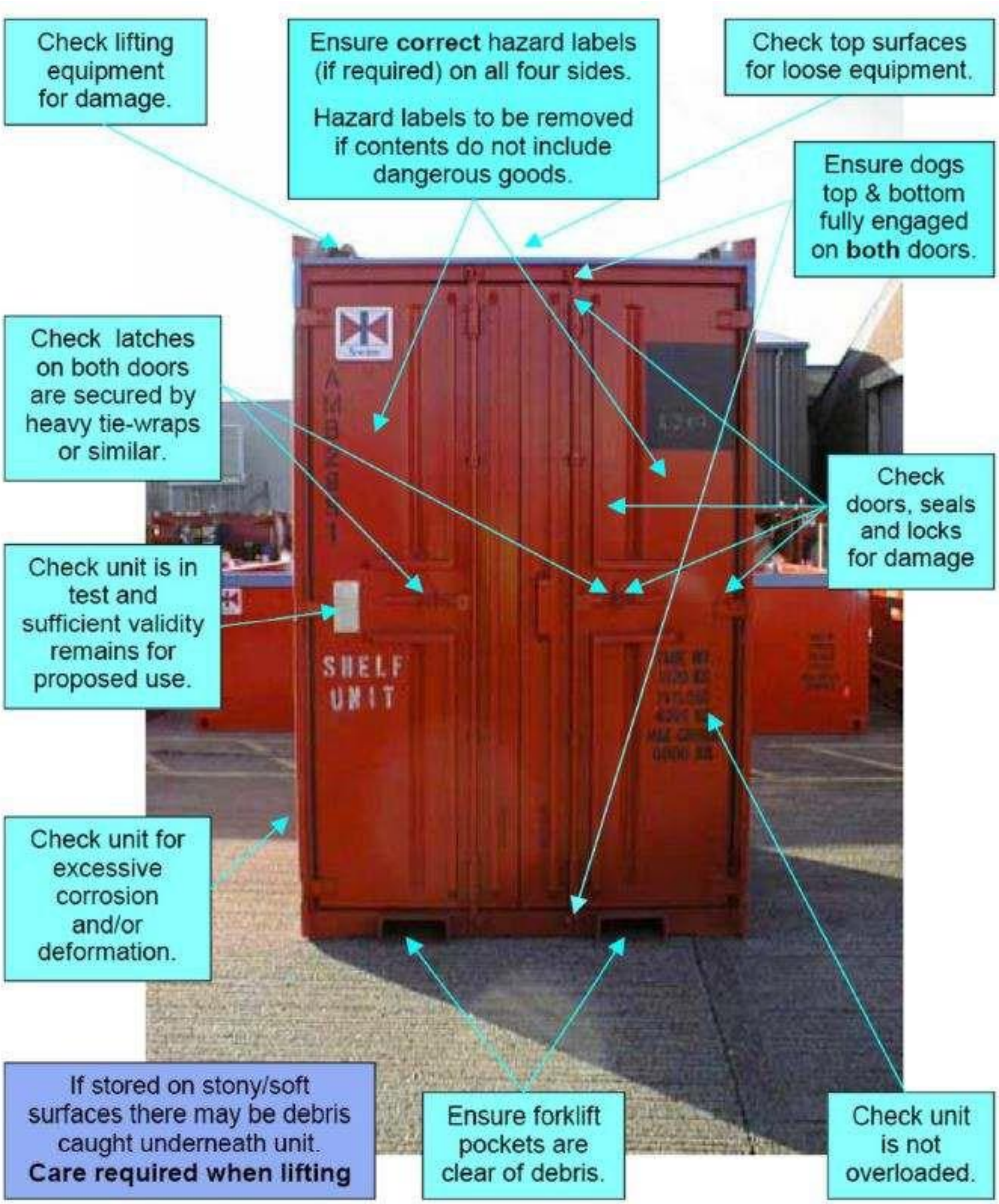
QA/QC Flowchart



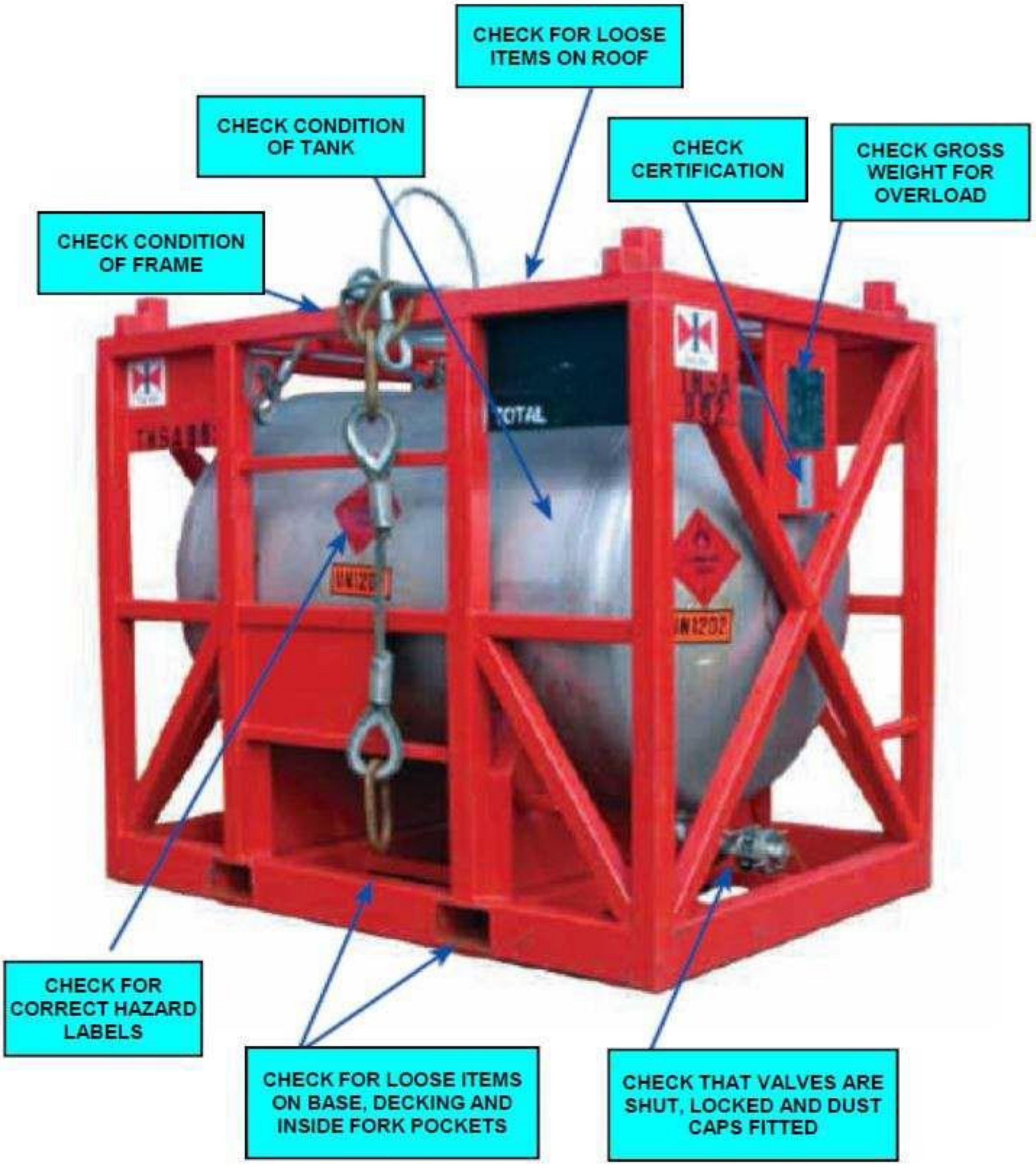
QA/QC Flowchart
Examination of Sling Sets & Shackles

Wire Rope Sling Sets
Examine each individual leg along its entire length and check for: 1) Wear / Corrosion 2) Abrasion 3) Mechanical Damage 4) Broken Wires Examine each ferrule and ensure the correct size of ferrule has been fitted. Check that the end of the loop does not terminate inside the ferrule (i.e. the rope end should protrude slightly but no more than 1/3rd of the dia.) unless the ferrule is of the longer tapered design which has an internal step. The ferrule should be free from cracks or other deformities. Examine each thimble and check for correct fitting, snagging damage and elongation. (Stretched thimbles / eyes could indicate possible overload). Examine wire rope around thimbles as it is often abraded due to sling being dragged over rough surfaces. If fitted, examine master link / quadruple assembly and check for wear, corrosion and cracking. If fitted with hooks, check for wear, corrosion and cracking and ensure safety latch functions.
Shackles
Check that the SWL is adequate for the load. Check that the color-coding is current and the shackle has a plant number / ID mark. Examine shackle body and check for wear in the crown and pinholes, deformation and cracking. In the case of safety pin shackles, ensure split pins are fitted. Refer to " Safe Cargo Handling " Pocket Book for further reference.


Check top surfaces for loose equipment.
If loaded, ensure there are no potential internal snagging hazards. If so, ensure adequate protection is in place to prevent risk of snagging.
Check unit is in test and has sufficient validity remains for proposed use.
Check lifting equipment for damage.
Check doors and locks for damage.
Check unit is not overloaded
Check unit for excessive corrosion and/or deformation.
If stored on stony/soft surfaces there may be debris caught underneath unit. **Care required when lifting.**
Ensure drainage holes are clear.
Ensure forklift pockets clear of debris.
Check the locking mechanism is secured on both door handles.



Check lifting equipment for damage.
Ensure **correct** hazard labels (if required) on all four sides.
Hazard labels to be removed if contents do not include dangerous goods.
Check top surfaces for loose equipment.
Ensure dogs top & bottom fully engaged on **both** doors.
Check latches on both doors are secured by heavy tie-wraps or similar.
Check doors, seals and locks for damage
Check unit is in test and sufficient validity remains for proposed use.
SHELF UNIT
Check unit for excessive corrosion and/or deformation.
If stored on stony/soft surfaces there may be debris caught underneath unit.
Care required when lifting
Ensure forklift pockets are clear of debris.
Check unit is not overloaded.


CHECK FOR LOOSE ITEMS ON ROOF
CHECK CONDITION OF TANK
CHECK CERTIFICATION
CHECK GROSS WEIGHT FOR OVERLOAD
CHECK CONDITION OF FRAME
CHECK FOR CORRECT HAZARD LABELS
CHECK FOR LOOSE ITEMS ON BASE, DECKING AND INSIDE FORK POCKETS
CHECK THAT VALVES ARE SHUT, LOCKED AND DUST CAPS FITTED

APPENDIX 8 - MOBILE CRANE ENTRY CHECKLIST

This form to be completed before crane enters site		
Name of Crane Company:		
Type of Crane:		
Crane capacity:		
Registration Number:		
Serial Number:		
I confirm that I have copies of the following statutory Documents for the crane:		
Proof Load Test Certificate		Checked (tick)
12 Monthly Thorough Examination Inspection Report		
If Man-Riding involved, 6 Monthly Examination Certificate Mandatory		
6 Monthly Thorough Examination Inspection Report for Slings and Lifting Tackle Supplied with crane		
Weekly Inspection Report		
Copies of the “Drivers / Operator Training” or Achievement Test Certificate		
Copies of the above-mentioned Documents must be available for inspection by the Site Supervisor		
Checked / Verified by		
Name (print):		
Signature	Date:	

APPENDIX 9 - FORKLIFT ENTRY CHECK SHEET

This form is to be completed before forklift enters site	
Name of Hire Company:	
Type of Forklift:	
Forklift capacity:	
Registration Number:	
Serial Number:	
I confirm that I have copies of the following statutory Documents for the forklift:	

Proof Load Test Certificate	**Checked (tick)**
12 Monthly Thorough Examination Inspection Report	
If Man-Riding involved, 6 Monthly Examination Certificate Mandatory	
6 Monthly Thorough Examination Inspection Report for any attachments such as jib attachments, hooks etc.	
Weekly Inspection Report	
Copies of the “Drivers / Operator Training” or Achievement Test Certificate	

Copies of the above-mentioned Documents must be available for inspection by the Site Supervisor. Checked / Verified by Name (print):
Signature:
Date:

APPENDIX 10 - DAILY OPERATORS PRE-USE INSPECTION ELECTRIC FORKLIFT

This form to be completed daily before use

Date Vehicle No. Type and Model Hour Meter

Visual Checks	**OK**	**NA**	**Maintenance Needed** **Reported To:**
Leaks – Hydraulic Oil, Battery			
Tires – Condition and pressure			
Forks, Top Clip retaining pin and heel – Condition			
Load Backrest Extension – Solid attachment			
Hydraulic hoses, mast chains and stops			
Finger guards - Attached			
Safety Warnings – Attached and legible			
Operators Manual – Located on truck and legible			
Capacity Plate – Attached: information matches Model & Serial Nos. and attachments.			
Seat Belt – Buckle and retractor working smoothly			
Accelerator Linkage			
Parking Brake / Deadman			
Steering			
Drive Control – Forward and Reverse			
Tilt Control – Forward and Back			
Hoist & Lowering Control			
Attachment Control			
Horn			
Lights			

Back-Up Alarm			
Hour Meter			
Battery Discharge Gauge			
Fire Extinguisher			
Inspected by: Custodian:			

APPENDIX 11 - DAILY OPERATORS PRE-USE INSPECTION DIESEL FORKLIFT

This form to be completed daily before use

Date Vehicle No. Type and Model
Hour Meter

Visual Checks	OK	NA	Maintenance Needed Reported To:
Fluid Levels – Oil Radiator, Hydraulic			
Leaks – Hydraulic Oil, Battery, Fuel			
Tires – Condition and pressure			
Forks, Top Clip retaining pin and heel –			
Condition			
Load Backrest Extension – Solid			
attachment			
Hydraulic hoses, mast chains & stops			
Finger guards - Attached			
Safety warnings – Attached and legible			
Operators Manual – Located on truck and			
legible			
Capacity Plate – Attached; information			
matches Model & Serial Nos and			
attachments			
Seat Belt – Buckle and retractor working			
smoothly			
Accelerator Linkage			
Parking Brake			
Steering			
Drive Control – Forward and Reverse			
Tilt Control – Forward and Back			

Hoist & Lowering Control			
Attachment Control			
Horn			
Lights			
Back-Up Alarm			
Hour Meter			
Fire Extinguisher			
Inspected by: Custodian:			

APPENDIX 12 - JUMBO BAG STANDARDS

1.SUMMARY

QatarEnergy-IS and its contractors all use Test Standard ISO 21898 Jumbo bags.

2.SCOPE

It applies to all Jumbo Bags used at all QatarEnergy-IS operated sites.

OPERATIONAL REQUIREMENTS

Jumbo bags should not be used for dynamic lifting i.e.to or from vessel to offshore platform.

Jumbo bags shall be protected from sunlight and moisture at storage areas. If bags have been exposed to sunlight for any extended period requires supervisor inspection and approval

Jumbo bags shall not be dragged or lifted by sharp edges.

Jumbo bag manufacturer's safe handling and stacking instructions shall always be followed.

Jumbo bags shall be certified as per the relevant standard.

CONTROL MEASURES

As a minimum requirement all jumbo bags complete with lifting straps shall be rated for the content weight and have a minimum Factor of Safety of 5:1. Bags should conform to BS EN ISO 21898

- Do NOT fill Jumbo bags to their 100% SWL capacity, maximum fill 75% of SWL capacity.
- Jumbo bags shall not be reused or re-circulated. Once emptied bag shall be discarded/destroyed accordingly.
- Jumbo bags shall have four (4) lifting points from lifting straps that are completely encircling the bag.
- Jumbo bags shall be lifted by using a four-leg sling/bridle (Spreader Bar if required). Bag shall not be lifted by a single leg sling or a single loop.

A batch testing Procedure as detailed below shall be followed for certification of the bags.

1. All bags shall be marked with an individual ID number,
2. The break test certificate number shall be referred on the certificate issued for the remaining bags (with unique ID nos.),
3. The details such as ID no., SWL and Break test certificate number shall be recorded in each certificate.

ADDITIONAL CONTROL MEASURES

Do's	Don't
Do read the instruction label on the Jumbo bag	Don't exceed the SWL in any circumstances
Do ensure that the filled Jumbo bag is stable	Don't subject Jumbo bags to snatchlift and/or jerking
Do use lifting gear of sufficient capacity to take the suspended load	Don't allow personnel under suspended Jumbo Bags
Do take appropriate measures with regard to dust control	Don't stack Jumbo bags unless sure of their stability
Do protect the Jumbo bags from rain and/or prolonged	Don't reuse single-trip Jumbo bags
Do ensure the Jumbo bags are adequately secured in transportation	Don't transport Jumbo bags unless they are inside a CCU (Cargo Carrying Unit)

INSPECTING JUMBO BAGS

Jumbo bags shall be thoroughly examined by a competent person for damage to stitching and for surface abrasion, cuts, tears or any other damage to the bag. Particular attention should be paid to the lifting loops or devices and their attachments. The examination should look for signs of the following.

1. Abrasion
2. Cuts
3. Damage to coatings
4. Ultraviolet degradation, distortion and/or chemical attack

If and of these signs are visible then the bag shall not be used and must be discarded accordingly.

MARKING OF JUMBO BAGS

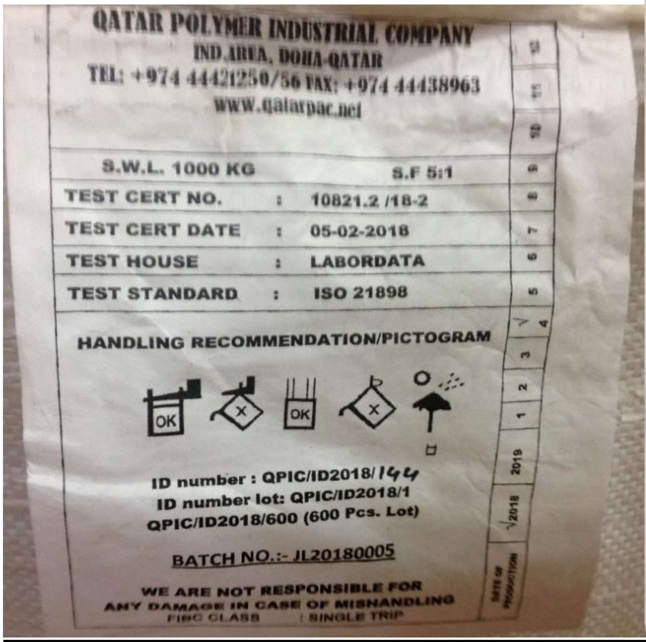
All Jumbo bags shall be durably marked by means of a permanently attached and easily visible and readable label, or durably printed on the body so that it is easily visible and read after the Jumbo bag has been filled. The following data shall be included:

- name and address of the manufacturer;
- manufacturer's reference, which shall be unique to any one FIBC type;
- name and address of the supplier, if required;
- safe working load (SWL) in kilograms;
- safety factor (SF), i.e. 5:1, 6:1 or 8:1 as appropriate;
- reference to International Standard ISO 21898
- class of Jumbo bag, i.e. "heavy-duty reusable", "standard-duty reusable" or "single-trip";
- type test certificate number (which shall be unique to any one type) and the month and year in which the type test certificate was issued:
- name of the approved laboratory;
- date of manufacture of the Jumbo bag, i.e. month and year;
- pictograms of the recommended handling methods;

STORAGE OF JUMBO BAGS

- Empty Jumbo bags should be stored in such a manner that accidental damage, exposure to sunlight or extreme climatic conditions, and contact with substances likely to degrade the materials are avoided.
- Storage of filled Jumbo bags at temperatures above 50 °C should be avoided, except with the approval of the manufacturer or supplier.
- Filled Jumbo bags should have any top closures properly closed before storage.
- All Jumbo bags stored outdoors should, be sheeted over to prevent water collection on the tops of FIBCs, not be stored in standing water, and be protected against rays of sunshine.

CERTIFICATION EXAMPLES


QATAR POLYMER INDUSTRIAL COMPANY
IND.AREA, DOHA-QATAR
TEL: +974 44421250/56 FAX: +974 44438963
www.qatarpac.net

S.W.L. 1000 KG		S.F 5:1
TEST CERT NO.	:	10821.2 /18-2
TEST CERT DATE	:	05-02-2018
TEST HOUSE	:	LABORDATA
TEST STANDARD	:	ISO 21898

HANDLING RECOMMENDATION/PICTOGRAM

OK OK

ID number : QPIC/ID2018/144
ID number lot: QPIC/ID2018/1
QPIC/ID2018/600 (600 Pcs. Lot)

BATCH NO.:- JL20180005

WE ARE NOT RESPONSIBLE FOR
ANY DAMAGE IN CASE OF MISHANDLING
FIBC CLASS : SINGLE TRIP

DATE OF PRODUCTION 2018 2019 1 2 3 4 5 6 7 8 9 10 11 12


شركة قطر بوليمر الصناعية
Qatar Polymer Industrial Company
Manufacturers of PP Woven Sacks, PE Liners and FIBC

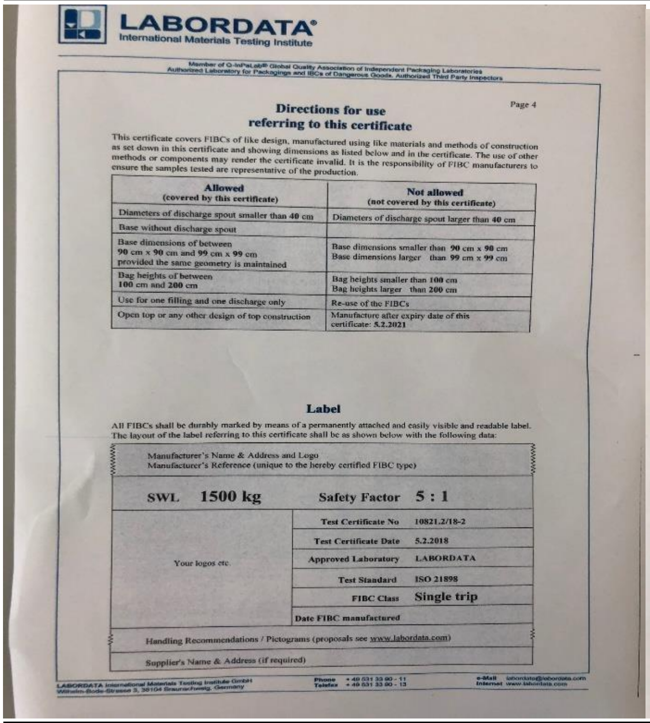
EC CERTIFICATE OF CONFORMITY

CERTIFICATE NUMBER : 5877 DATE: 22.07.2018

SL NO.	PARAMETER	FINDINGS
1	Bag Dimension	90cm x90cm x 120cm, Colour: White
2	Safe working load	1500Kg
3	Safety factor	5:1
4	Manufacter date	Jul-18
5	Type of bag	Circular cross corner full loops
6	Lot No.	(100 pcs)
7	Batch No.	JB20182626
8	Single /Multi use	Single use
9	Unique ID No.	ID number lot: QPIC/ID2018/1 to QPIC/ID2018/100
10	Test Certificate number	10821.2/18-2
11	Test Certificate Date	05.02.2018
12	FIBC Class	Single trip
13	Test standard	BS EN ISO 21898 : 2005 /QP REG.Q 001 REV.04
14	Approved Laboratory	Labordata , Germany
15	Lifting Mathod	Using forklift 4-point lifting ,two loops on one fork and other two loops on the one fork of the forklift
16	Loops	Full Loop
17	Top style	Open
18	Bottom	Flat
19	PE Liner Bag	No

P.O. Box: 33195, Doha - Qatar, Tel.: +974 44421250/40372463 Fax: +974 44438963, C.R. No.: 19314, Paid-up Capital (QR): 106,081,610
Email: info@qatarpac.net, Website: www.qatarpac.net,
An ISO 9001:2015, OHSAS 18001:2007 & HACCP Certified Company
We bring packaging to life

Cert No. 14878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Authorized Laboratory for Packagings and IBCs of Dangerous Goods. Authorized Third Party Inspectors

Page 4

Directions for use referring to this certificate

This certificate covers FIBCs of like design, manufactured using like materials and methods of construction as set down in this certificate and showing dimensions as listed below and in the certificate. The use of other methods or components may render the certificate invalid. It is the responsibility of FIBC manufacturers to ensure the samples tested are representative of the production.

Allowed (covered by this certificate)	Not allowed (not covered by this certificate)
Diameters of discharge spout smaller than 40 cm	Diameters of discharge spout larger than 40 cm
Base without discharge spout	
Base dimensions of between 90 cm x 90 cm and 99 cm x 99 cm provided the same geometry is maintained	Base dimensions smaller than 90 cm x 90 cm Base dimensions larger than 99 cm x 99 cm
Bag heights of between 100 cm and 200 cm	Bag heights smaller than 100 cm Bag heights larger than 200 cm
Use for one filling and one discharge only	Re-use of the FIBCs
Open top or any other design of top construction	Manufacture after expiry date of this certificate: 5.2.2021

Label

All FIBCs shall be durably marked by means of a permanently attached and easily visible and readable label. The layout of the label referring to this certificate shall be as shown below with the following data:

Manufacturer's Name & Address and Logo
Manufacturer's Reference (unique to the hereby certified FIBC type)

SWL 1500 kg Safety Factor 5 : 1

Your logos etc.	Test Certificate No	10821.2/18-2
	Test Certificate Date	5.2.2018
	Approved Laboratory	LABORDATA
	Test Standard	ISO 21898
	FIBC Class	Single trip
	Date FIBC manufactured	

Handling Recommendations / Pictograms (proposals see www.labordata.com)

Supplier's Name & Address (if required)

LABORDATA International Materials Testing Institute GmbH, Wilhelm-Bode-Strasse 3, 38104 Braunschweig, Germany — Phone +49 531 33 90 - 11, Telefax +49 531 33 90 - 13 — e-Mail labordata@labordata.com, Internet www.labordata.com


LABORDATA®
International Materials Testing Institute

Member of Q-InPaLab® Global Quality Association of Independent Packaging Laboratories
Authorized Laboratory for Packagings and IBCs of Dangerous Goods. Authorized Third Party Inspectors

Date of test 5.2.2018
Date of expiry 5.2.2021
Number of pages 4 C / D
This Certificate is only valid when printed in colour and complete with all 4 pages.

Test Certificate No. 10821.2/18-2

Applicant Qatar Polymer Industrial Company, P.O. Box: 33195, Doha, Qatar

Test piece *Flexible Intermediate Bulk Container - SWL = 1500 kg, SF = 5:1*
Single trip FIBCs for non-dangerous goods acc. ISO 21898

Design
Manufacturer's type designation QPIC - CRZZ - 1500 KG 5:1 / 2018
Dimensions: Sample a : (90 cm x 90 cm) x 100 cm (outer size) Volume 900 litres Tare 1420 g; Samples b + c: (90 cm x 90 cm) x 200 cm (outer size) Volume 1800 litres Tare 2020 g
Wall fabric: Circular fabric. Polypropylene 180 g/m², uncoated (average incl. reinforcing stripes), white fabric with four red and twenty blue coloured tapes and eight interwoven reinforcing stripes
Base fabric: Polypropylene 190 g/m², uncoated
Suspension: Four white PP-webbings (75 mm wide, 48 g/m), sewn flat onto the reinforcing stripes in a length of 45 cm / 45 cm, cross corner design
Details: No vertical seams, four horizontal seams at the bottom (overlock + chain stitching) / wall and base fabric folded in the bottom seams / open top / no inliner / discharge spout d = 40 cm made of PP-fabric 70 g/m² + 20 g/m² coating, double seam

Kind of tests *Type Tests according ISO 21898*
Tests a + b: Cyclic top lift tests plus final load to failure. Test c: Compression test

Test conditions Charging with plastic granules (filling height approx. 95 cm (lowest size) resp. 195 cm (highest size). load application with piston and pressure plate (d = 90 cm), rate of load application 70 kN/min.

Cyclic load and load to failure
Sample a: After 30 cycles of load application to P = 30 kN (3060 kg) no visible damages occurred in the test piece. The load has then been increased until failure. On reaching a load of P = 74,3 kN (7570 kg) the base fabric tore at a bottom seam.
Sample b: After 30 cycles of load application to P = 30 kN (3060 kg) no visible damages occurred in the test piece. The load has then been increased until failure. On reaching a load of P = 85,8 kN (8740 kg) the wall fabric tore horizontally below two webbing attachments.

Compression Sample c: After six hours compression by P = 60 kN (6120 kg) no visible damages occurred in the test piece.

Test result *A safe working load SWL = 1500 kg / SF = 5:1 is allowable.*

Statement of conformity The FIBCs tested comply with the requirements of ISO 21898. FIBCs of this design type are in a condition for safe operation.

Notes
This Certificate is restricted to FIBCs produced by Qatar Polymer Industrial Company.
This certificate covers all FIBCs with heights of between 100 cm and 200 cm.
All material weights are minimum weights and may not be lower than the values shown.
Test diagrams see page 2. Photos of the test pieces see page 3.
Raw material: Pure virgin polypropylene (statement of the manufacturer)
"Directions for use referring to this certificate" see page 4.
Two test pieces are kept in our store for three years. This certificate expires on 5.2.2021.

Competent Engineer: Ronald Clews
Head of Institute: Dr. Herbert Kielbassa

LABORDATA International Materials Testing Institute GmbH, Wilhelm-Bode-Strasse 3, 38104 Braunschweig, Germany — Phone +49 531 33 90 - 11, Telefax +49 531 33 90 - 13 — e-Mail labordata@labordata.com, Internet www.labordata.com

APPENDIX 13 - REVISION HISTORY LOG

Revision Number: 03 **Document Revision Date: 03/09/2024**

Item Revised	Revision Description	Page No.
Miscellaneous	• Updated Purpose of Document page 3 • Minor updates to Roles and Responsibilities page 3 • Deleted 4.1 Planning Risk Assessment, competence section page 6 • Deleted 4.1.1 Planning section A and B page 6 • Deleted 4.1.2 Training and competence page 7 • Deleted section 4.2 Safe use of lifting and handling equipment page 8 • Deleted section 4.2.4 Mobile crane requirements page 11 • Minor updates to 4.2.6.1 Man riding winches to include PTS stamp requirement page 13 • Deleted section 4.2.6.3 Cranes used for man riding operations page 15 • Deleted section 4.4 Written Scheme of examination page 17 • Updated 5.0 References to include both corporate procedures • Deleted Appendix 2 planning complex and complicated lifts page 26 • Deleted Appendix 3 QatarEnergy IS lifting color code page 30 • Deleted Appendix 6 Approved third party lists page 35 • Deleted Appendix 3 Deviation form page 38 • Deleted Appendix 8 Lifting equipment matrix page 40	3, 6, 7, 8, 11, 13, 15, 17, 26, 30, 35, 38, 40
Remarks : Revised to align with corporate requirements.		

APPENDIX 14 – LIST OF ABBREVIATION

Term	Abbreviation
CCU	Cargo Carrying Unit
FOS	Factor of Safety
FRC	Fast Rescue Craft
MBL	Minimum Breaking Load

MODU	Mobile Offshore Drilling Unit
MOU	Mobile Offshore Unit
NCR	Non-conformance Report
NDT	Non-destructive testing
PIC	Person in Charge
PLT	Proof Load Test
SWL	Safe working load
TPCA	Third Party Certification Authority
WHJ	Wellhead Jacket
WLL	Working Load Limit
ERRV	Emergency Response Rescue Vessel